



XMBA 200S: Data and Decisions

Chapter 13: Samples and Surveys

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Data and Decisions Part 1: The data you want versus the data you've got

- Managers (and others) almost never have all of the data they want . . .
 - On their customers
 - On their production facilities
 - On their employees' productivity
- Or, if they do have all the data they want, perhaps the data is so large as to be unwieldy
- If you can only collect **a sample of data**
 - **Today:** How should you select that sample?
 - **Next:** What can you say about the overall “population”, and how precisely can you say it?

What we'll discuss today

- Two crucial issues in choosing and using samples:
 1. Sampling variation – your answers won't be exactly right, but **statistics can tell us how close we are to being right**
 2. Bias – your answers may **systematically** be wrong
- Survey implementation
 - SRS, Stratified, Clustered samples
 - Why randomization is great; why relatively small samples from large populations are ok
 - Why lack of precision is tolerable, but bias is not

J.D. Power and Associates



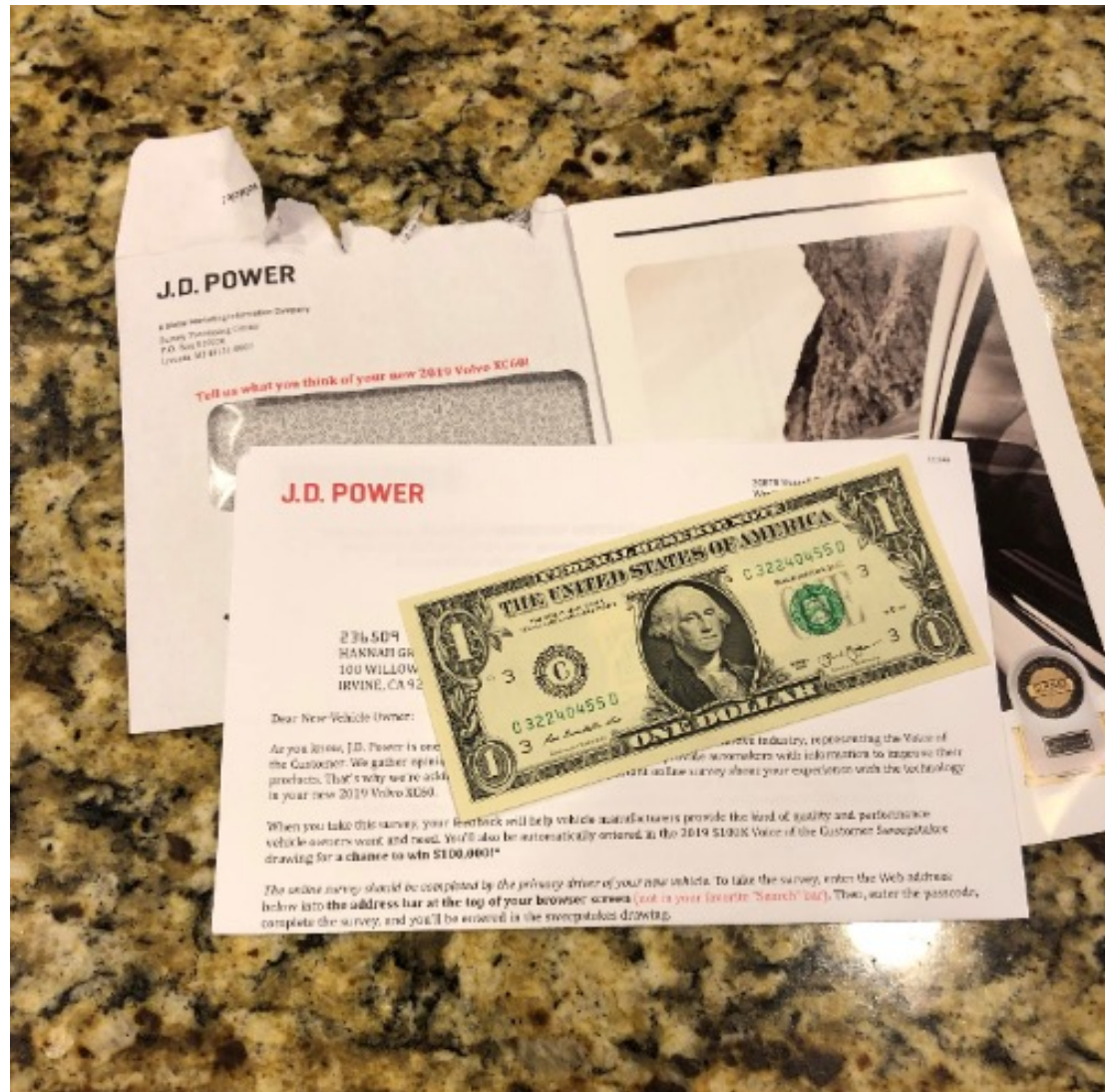
- A global marketing information services firm.
- Company specializes in customer satisfaction. Initial Quality Study & Vehicle Dependability Study (VDS)
- Surveys are run annually. Data production company.
- Best known product is the “Vehicle Dependability Study”.
What is this? Where do these data come from?

J.D. Power and Associates



- According to JD Power & Associates: **2025 U.S. Vehicle Dependability Study** is based on responses from 33,268 original owners of 2023 model-year vehicles after 3 years of ownership
- The survey for 2026 was fielded in 2025
- It measures the number of problems per 100 vehicles (PP100) experienced during the past 12 months by original owners of their three-year-old vehicles
- A lower score reflects higher quality, and the study covers 184 specific problems, grouped into 9 major vehicle categories

Survey Incentives

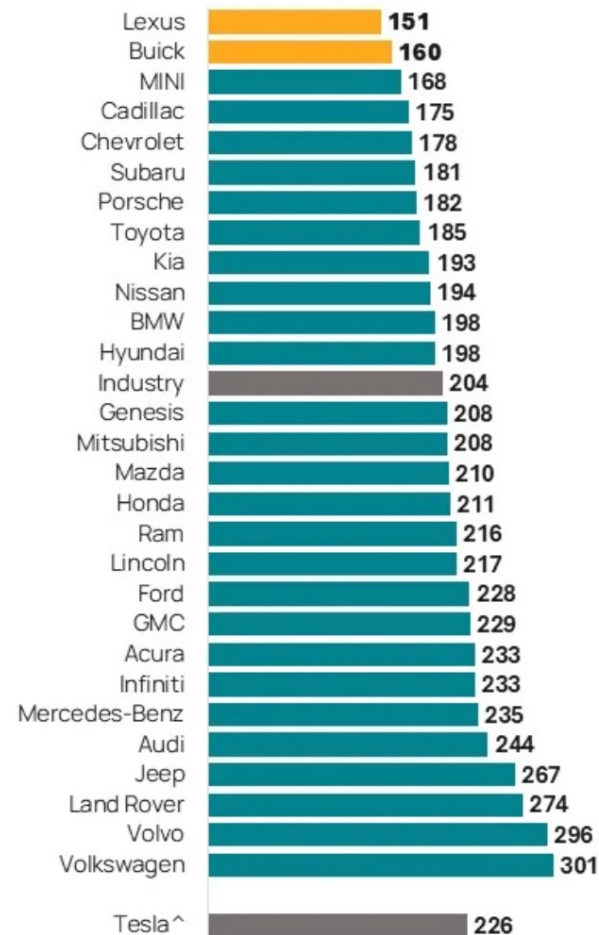


Vehicle Dependability Study

JD Power 2026 U.S. Vehicle Dependability Study (VDS)SM

Brand Ranking

Problems per 100 Vehicles (PP100)



Definitions

Population: the entire collection of interest

Census: a comprehensive survey of the entire population

Sample: subset of the population

Survey: posing questions to a sample to learn about the population

Representative: samples that reflect the mix in the entire population

Bias: systematic error in selecting the sample

Sentence of the Day:

“There is one population, but there are many different samples”

Eliminating Bias

- The best way to get a representative sample is to pick members of the population at random.
 - A random sample is representative of the whole population.
 - Randomization eliminates bias.
 - Randomization ensures that on average a sample mimics the population.

How to Randomize

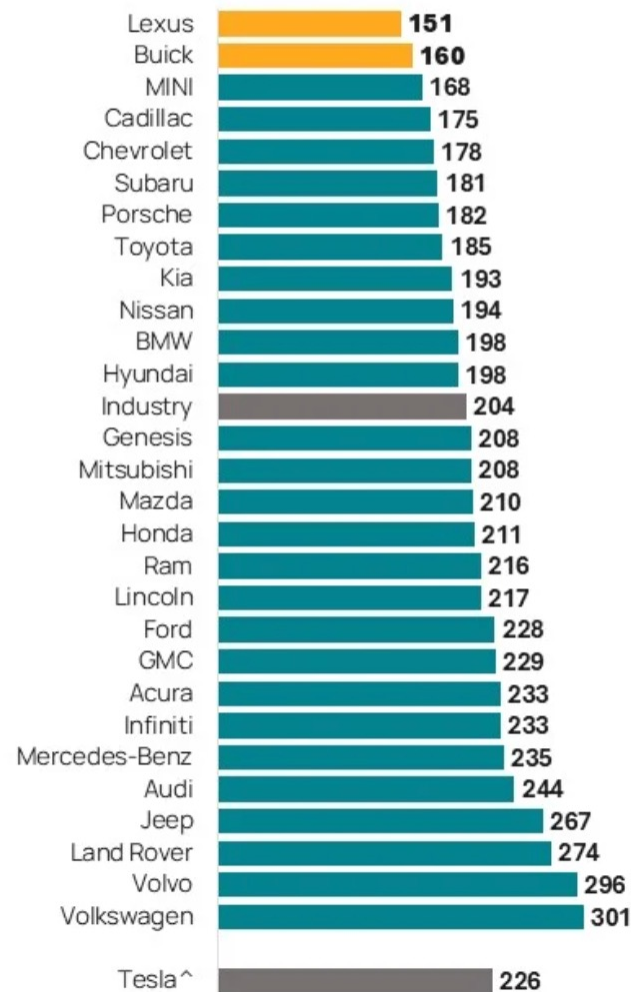
- **Simple Random Sample (SRS):**
 - A sample of size n is chosen randomly from the population.
 - The SRS is the gold standard to which all other sampling methods are compared.
- **Random Number Generator:**
 - draw for each item a number from the uniform distribution $[0,1]$. If you want a 10% sample, for example, select all items with number $<.10$.

What might be some problems with a simple random sample?

JD Power 2026 U.S. Vehicle Dependability Study (VDS)SM

Brand Ranking

Problems per 100 Vehicles (PP100)



Alternatives to Simple Random Sample

- **Stratified Sample:**
 - a sample derived from random sampling within subsets of similar items known as strata
- **Easy Two Step Process:**
 1. Divide the sampling frame into strata
 2. Use simple random sampling to select items from each strata

Why might J.D. Power want to use a stratified sample?

Alternatives to Simple Random Sample

- **Cluster Sample:**
 - A type of stratified sampling that groups the population into conveniently surveyed geographic clusters
- **Easy Three Step Process:**
 1. Divide the population into geographic clusters
 2. Randomly select clusters
 3. Randomly choose items within selected clusters

*Use it when the area to survey is large & the cluster areas relatively homogeneous. This way you can focus on smaller subregions. **Cost savings** can be substantial relative to simple random sampling.*

Example of Cluster Random Sample: CPI

Bureau of Labor Statistics: Consumer Price Index

Cluster randomized design

- 1. Randomly pick geographic markets*
- 2. Randomly choose stores within each market*
- 3. Collect price data from representative “basket”*

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Consumer Price Index

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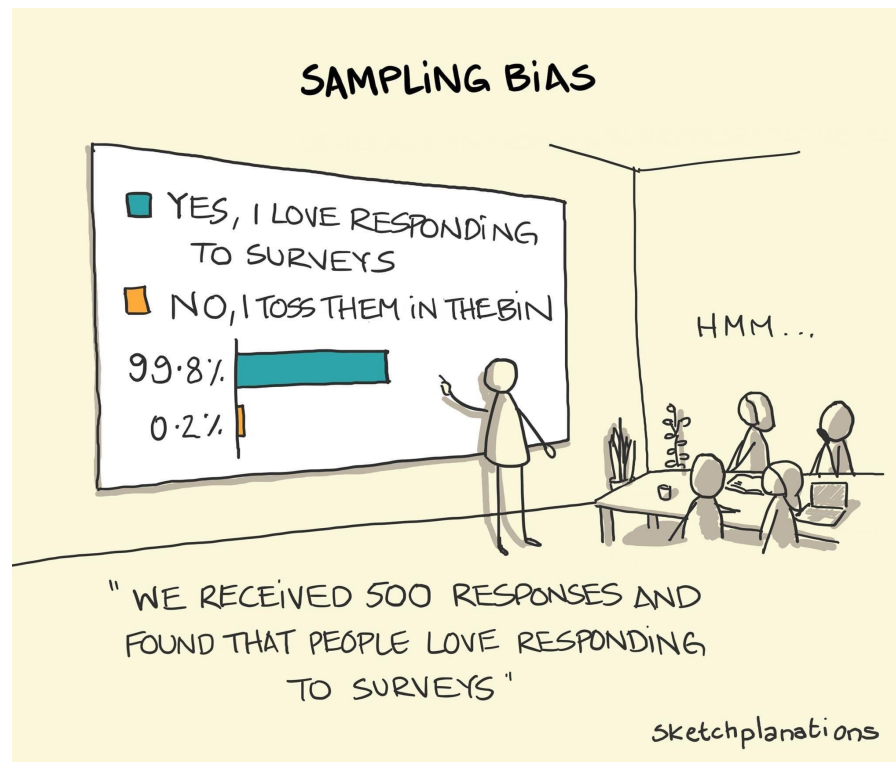
[Contact CPI](#)

Notice of CPI collection reductions

BLS is reducing sample collection in areas across the country. In April, BLS suspended CPI data collection entirely in Lincoln, NE, and Provo, UT. In June, BLS suspended collection entirely in Buffalo, NY.

Alternatives to Simple Random Sample

- **Voluntary Response Sample:**
 - This is a sample consisting of individuals who volunteer to participate in a survey.
 - The big challenge with these surveys is that they are not representative.



Biased samples

- We **assume unbiased samples** in much of the rest of the course
- But sample bias is everywhere
- Two main issues:
 1. Selection into the sample
 2. Problems with the survey design

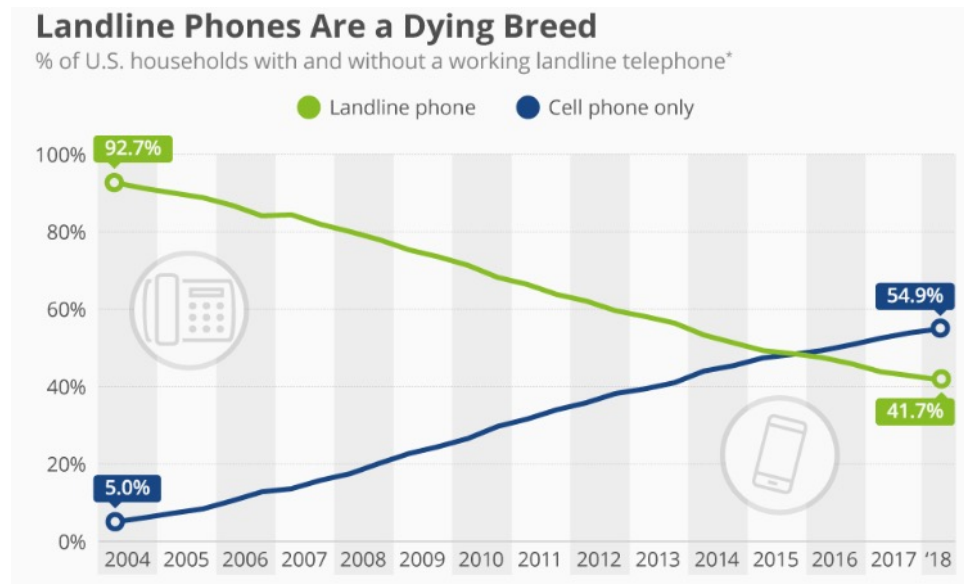
Samples with selection

Surveys Face Growing Difficulty Reaching, Persuading Potential Respondents

	1997	2000	2003	2006	2009	2012
	%	%	%	%	%	%
Contact rate (percent of households in which an adult was reached)	90	77	79	73	72	62
Cooperation rate (percent of households contacted that yielded an interview)	43	40	34	31	21	14
Response rate (percent of households sampled that yielded an interview)	36	28	25	21	15	9

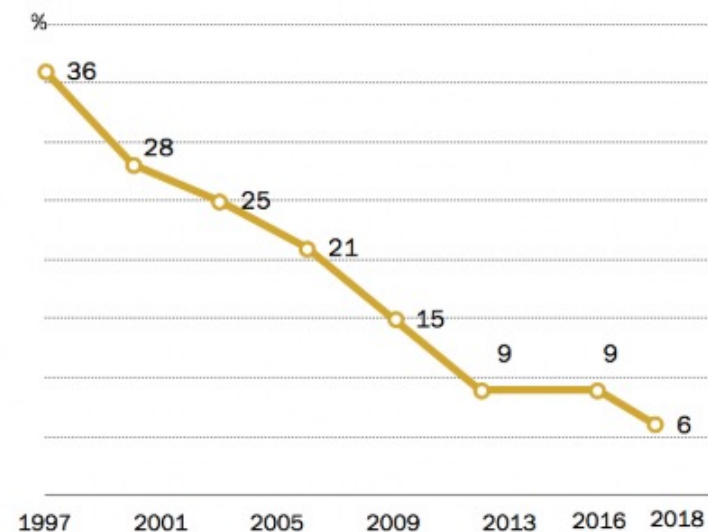
PEW RESEARCH CENTER 2012 Methodology Study. Rates computed according to American Association for Public Opinion Research (AAPOR) standard definitions for CON2, COOP3 and RR3. Rates are typical for surveys conducted in each year.

The Problem With Polling



After brief plateau, telephone survey response rates have fallen again

Response rate by year (%)



“The problem is simple but daunting. The foundation of opinion research has historically been the ability to draw a random sample of the population. That’s become much harder to do.”

--Nate Silver, quoted in the [New York Times](#)

Samples with selection

- The key question: who is left in; who is left out?
- If those left out look like those left in, you're ok
- This is why **random samples** are so valuable
 - Those included should be in no way systematically different from those excluded
 - Be careful of response bias
(i.e., non-random selection into survey)

Best Practices

As a data producer,

- ☐ Randomize.
- ☐ Match the sampling frame to the target population.
- ☐ Reduce nonresponse (e.g., financial incentives).

As a data consumer,

- ☐ Understand how sample was selected.
- ☐ Be skeptical when you see low response rates.

Inference

Now that we've discussed the strategies and challenges of sampling, let's move on to talk about statistical inference

Statistical inference: Using a sample statistic to learn about the underlying population parameter of interest

Estimating Population Parameters

- **Parameter:** a characteristic of the population
 - For example, the population mean, typically denoted μ
- **Statistic:** an observed characteristic of a sample
 - For example, the sample mean, typically denoted $\bar{x}$ or $\bar{y}$

Some notation:

Sample statistics vs. population parameters

- “It’s all Greek to me”

Name	Sample Statistic	Population Parameter
Mean	$\bar{y}$	μ (mu, pronounced “mew”)
Standard Deviation	s	σ (sigma)
Correlation	r	ρ (rho, pronounced “row”)
Slope of line	b	β (beta)
Proportion	$\hat{p}$	p

Sampling Variability

Sampling Variation: “statistics are noisy”

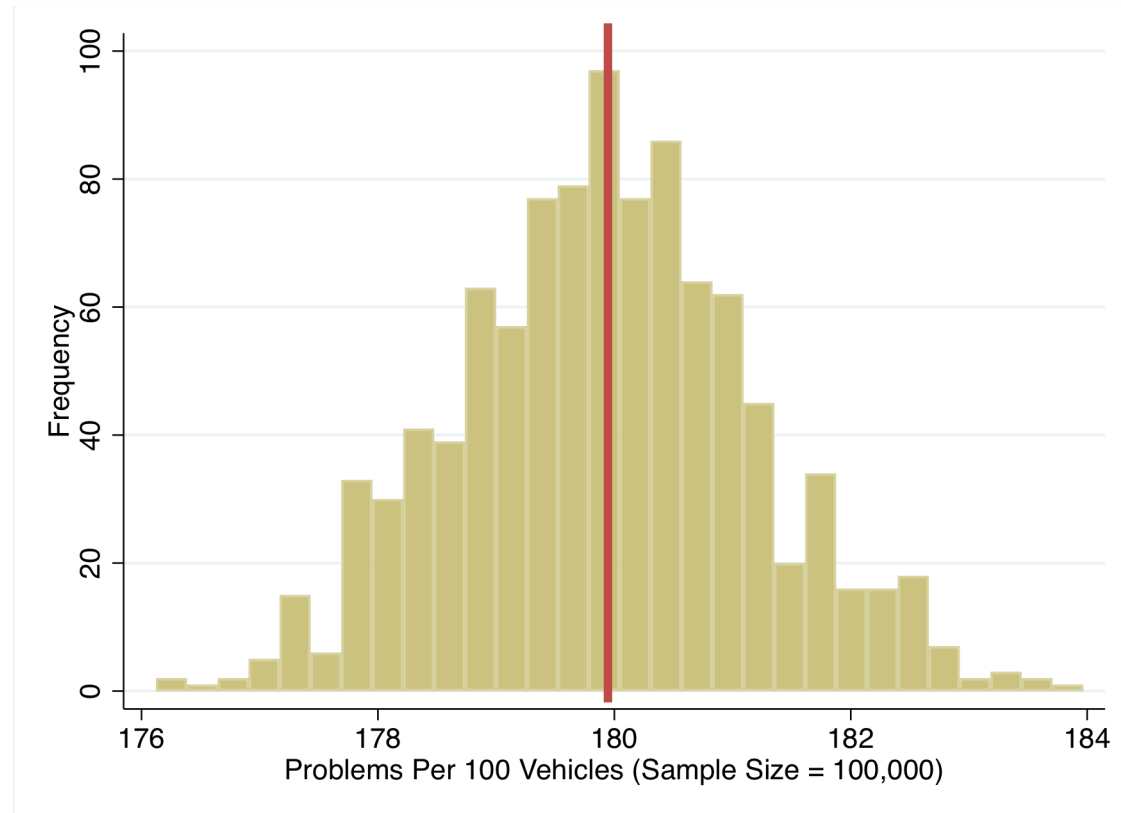
- variability in the value of a statistic from one sample to the next
- The price we pay for working with a sample rather than the population.

Sentence of the Day: Reminder

“There is one population, but there are many different samples”

Sampling Variability

The variability in a statistic from sample to sample.



Imagine JD Power did 1000 surveys from the same population and calculated a new sample mean from each of these 1000 surveys

Sampling Variability

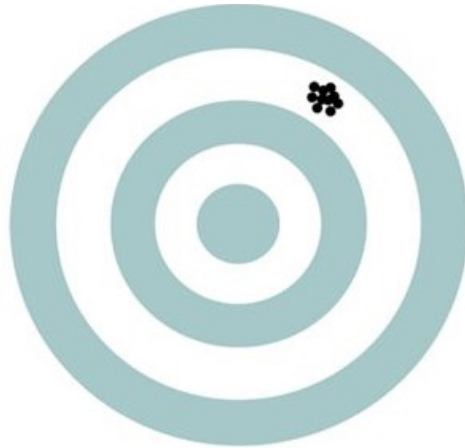
- Imagine flipping a coin ten times . . .
- If the coin is “fair”, how many times will heads come up?
- Let's do this!

Sampling Variability

- Suppose we did this again?
- And again?

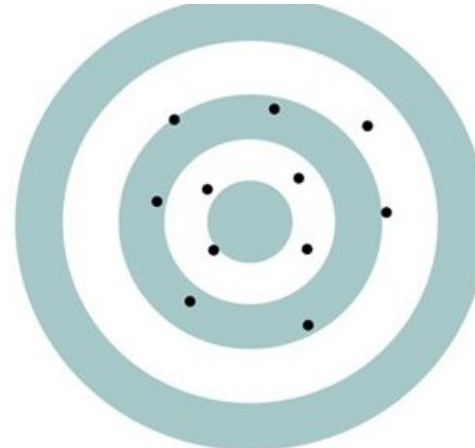
“Sampling variability” means strange things will be observed!

Bias versus Variability



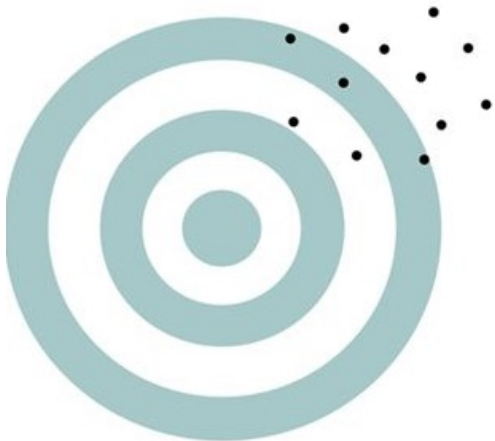
High bias, low variability

(a)



Low bias, high variability

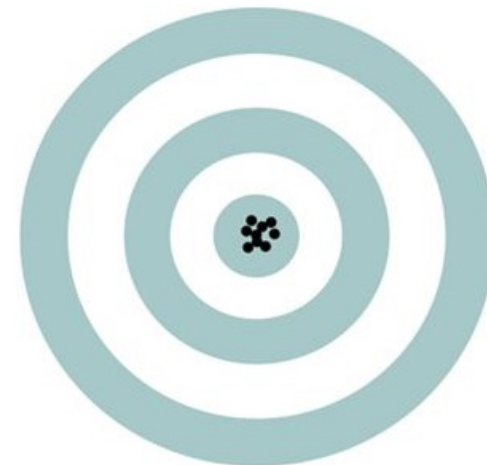
(b)



High bias, high variability

(c)

Properly chosen
statistics
computed from
random
samples of
sufficient size
will have low
bias and low
variability



The ideal: low bias, low variability

(d)

Sampling Variability

- The point: we rarely observe the exact population parameter we care about
- Thus, in the rest of the course, we'll use terms like "confidence"
 - We won't know things for sure, but we'll be able to learn quite a bit from **unbiased** samples

Punchline

- If you draw a potentially-biased (i.e., not representative) sample, what can you do?
- Run another survey
 - Try to do a better job collecting a random sample
 - Try to find observations you missed the first time
- Try to adjust your inferences to account for bias
 - We'll discuss tools to do this later in the course
 - “Bounds” or “Imputation” (link on bCourses)

Knowledge Check: Surveying to learn about (potential) customers

- Apple has a firmware/software update for their iPhone 17
 - How should they get this to customers? In store hardware/software update or automatic download?
 - Should they charge for the update? How much?

Spend a few minutes with your AI analyst thinking about how to conduct a sample/survey to get this information.